Topics to discuss

→ Simpson's 1/3 rd Rule

→ Numerical Problem

→ Error in Integration

→ Homework Problem

Simpson's $\frac{1}{3}$rd Rule :

It works by approximating the function with parabolic arcs instead of straight lines (like trapezoidal rule), which generally gives more accurate results.

The formula is :

$$\int_{x_0}^{x_n} y\,dx = \frac{h}{3}\left[y_0 + 4(y_1 + y_3 + y_5 + \cdots + y_{n-1}) + 2(y_2 + y_4 + \cdots\, y_{n-2}) + y_n\right].$$

Numerical Problem

Q: find the approximate value of $\int_0^1 \frac{x}{1+x^2}\,dx$ up to four decimal places by simpson 1/3 rd rule, taking 6 equal subintervals and find the approximate value of log 2.

Solution : length of interval will be,

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

Let, $f(x) = \frac{x}{1+x^2}$

We have, $f(x) = \frac{x}{1+x^2}$

x	0	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	1
$f(x)$	0	$\frac{6}{37}$	$\frac{3}{10}$	$\frac{2}{5}$	$\frac{6}{13}$	$\frac{30}{61}$	$\frac{1}{2}$

According to simpson's $\frac{1}{3}$rd rule.

$$\int_0^1 f(x)dx = \frac{h}{3}\left[y_0 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4) + y_6\right]$$

$$= \frac{1}{18}\left[0 + 4\left(\frac{6}{37} + \frac{2}{5} + \frac{30}{61}\right) + 2\left(\frac{3}{10} + \frac{6}{13}\right) + \frac{1}{2}\right]$$

$$= \frac{1}{18} \times (4.21586 + 1.52307 + 0.5)$$

$$= 0.3466$$ Ans.

Now, $f(x) = \frac{x}{1+x^2}$

$$\int_0^1 \frac{x}{1+x^2}\,dx$$

Let, $u = 1+x^2 \Rightarrow du = 2x\,dx \Rightarrow dx = \frac{du}{2x}$

So, $\int \frac{x}{1+x^2}\,dx = \int \frac{\cancel{x}}{u} \times \frac{du}{2\cancel{x}} = \frac{1}{2}\int \frac{du}{u} = \frac{1}{2}\ln|u| + C$

$= \frac{1}{2}\ln|1+x^2| + C$

Now,

$$\int_0^1 \frac{x}{1+x^2}\,dx = \left[\frac{1}{2}\ln|1+x^2|\right]_0^1 = \frac{1}{2}\ln 2$$

$\Rightarrow 0.3466 = \frac{1}{2}\ln 2$

$\Rightarrow \ln 2 = 0.6932$

$\therefore \log 2 = \frac{\ln 2}{\ln 10} = \frac{0.6932}{\ln 10} = 0.301$

Homework Problem

Q: Use Trapezoidal and simpson's $\frac{1}{3}$rd rule compute $\int_4^{5.2} \log_e x \, dx$ by taking seven ordinates correct up to four decimal places.

Solution:

Let, $f(x) = \log_e x$

$$\therefore h = \frac{b-a}{n} = \frac{5.2-4}{6} = 0.2$$

we have, $y = f(x) = \log_e x$

x	4	4.2	4.4	4.6	4.8	5.0	5.2
$f(x)$	1.3863	1.4351	1.48160	1.52605	1.56861	1.60943	1.64866

Now by trapezoidal rule,

$$\int_4^{5.2} f(x)\,dx = \frac{h}{2}\Big[1.3863 + 2\big(1.4351 + 1.48160 + 1.52605 + 1.56861 + 1.60943\big) + 1.64866\Big]$$

$$= \frac{0.2}{2} \times 18.27654 = 1.827654$$

Ans.

By simpson's $\frac{1}{3}$rd rule,

$$\int_{4}^{5.2} \log_e x \, dx = \frac{h}{3}\Big[1.3863 + 4(1.4351 + 1.5261 + 1.6094)$$

$$+ 2(1.4861 + 1.5686) + 1.6487\Big]$$

$$= 1.8279$$

Ans.

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